



DataDriven Floorplanning

ICCAD Problem C

B12901012 張維傑

B12901080 李俊翰

B12901185 謝承恩



Outline

- Legalization
- New Model
- GNN attempt

Legalization: Force directed

核心：6 個力同時作用（每步算一次合力，更新位置）

#	力	公式 / 行為	控制參數
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1	Repulsion	兩兩重疊沿較淺軸推開 $F = k_{rep} \cdot \min(overlap, cap)$	k_{rep} , $max_overlap_push$
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2	Anchor spring	每個 block 受朝向 $(cx0, cy0)$ 的彈簧力	k_{anchor}
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3	AR bias on push	決定推開軸時偏好讓整體 W:H 接近 1:1	$gamma$
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4	Frame wall	收縮中的矩形框邊界對溢出 block 推回	k_{frame} + band 感應
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5	Blank attraction	bbox 內 30×30 採樣，空白點對附近 block 用 $1/d^2$ 吸引	k_{blank} , $grid_n$
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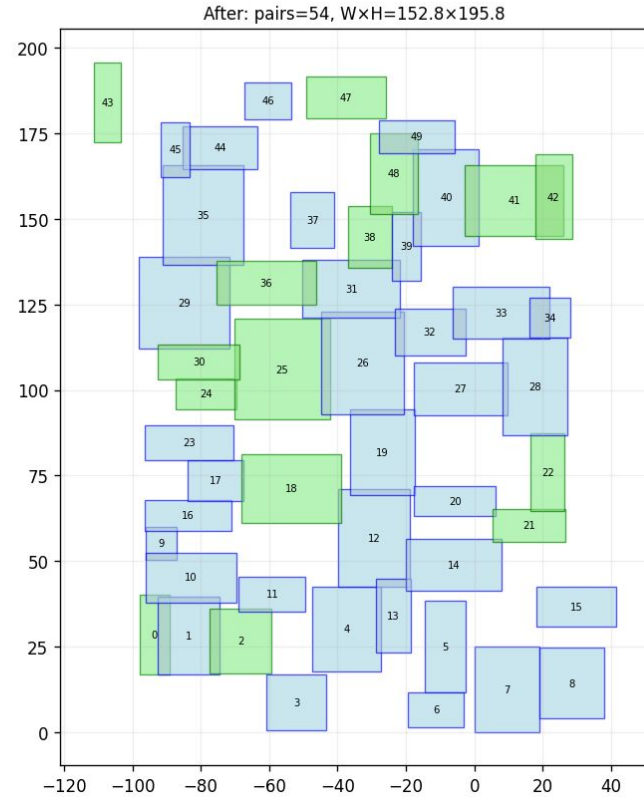
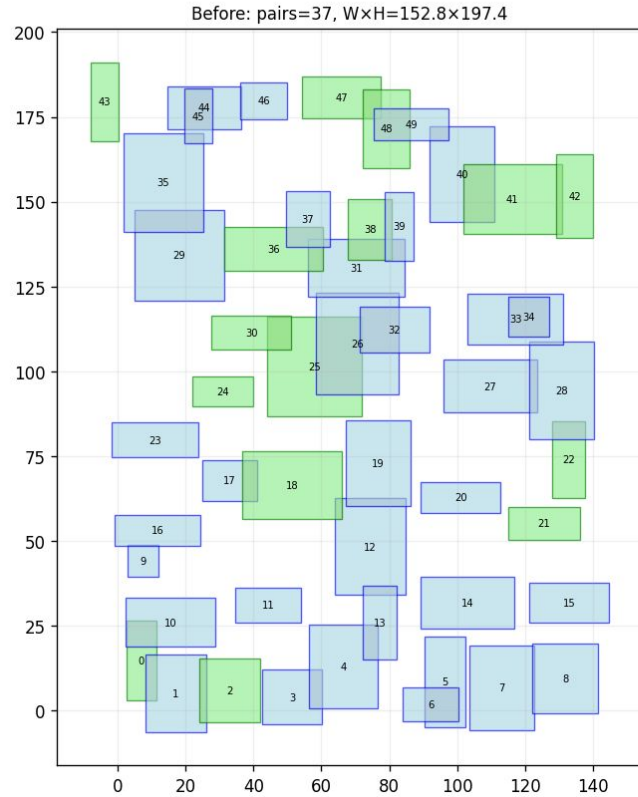
6	Rot90	機率性把方向不對的 block 翻 90°（中心不變）	rot_* 群
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Legalization: Force directed

- 沙子抖動嘗試：
 - 1.向中心靠攏的力
 - 2.隨著時間變化以中心點為原點右旋或左旋的力，模擬抖動一碗沙子

Legalization: test_01.csv

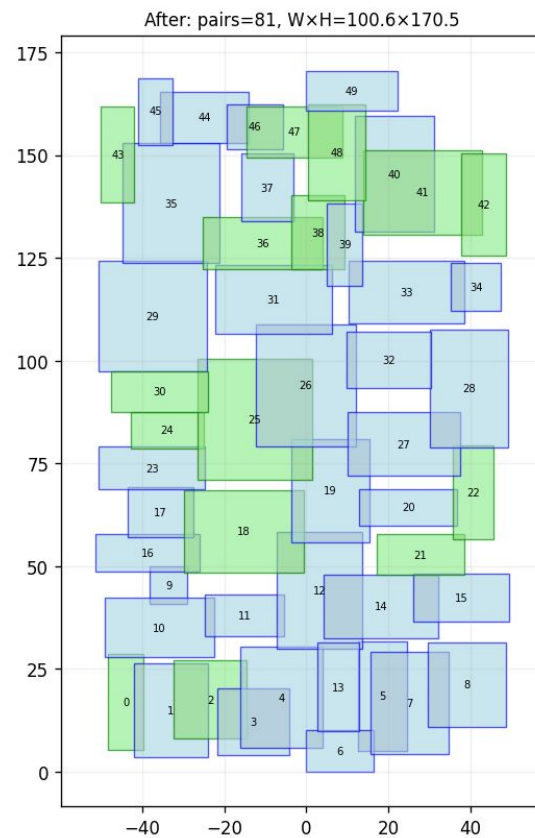
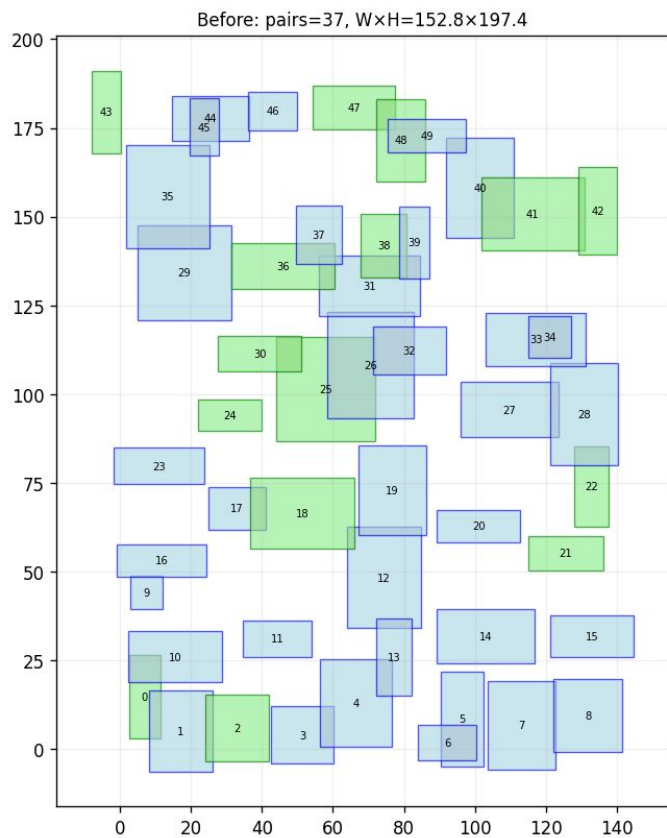




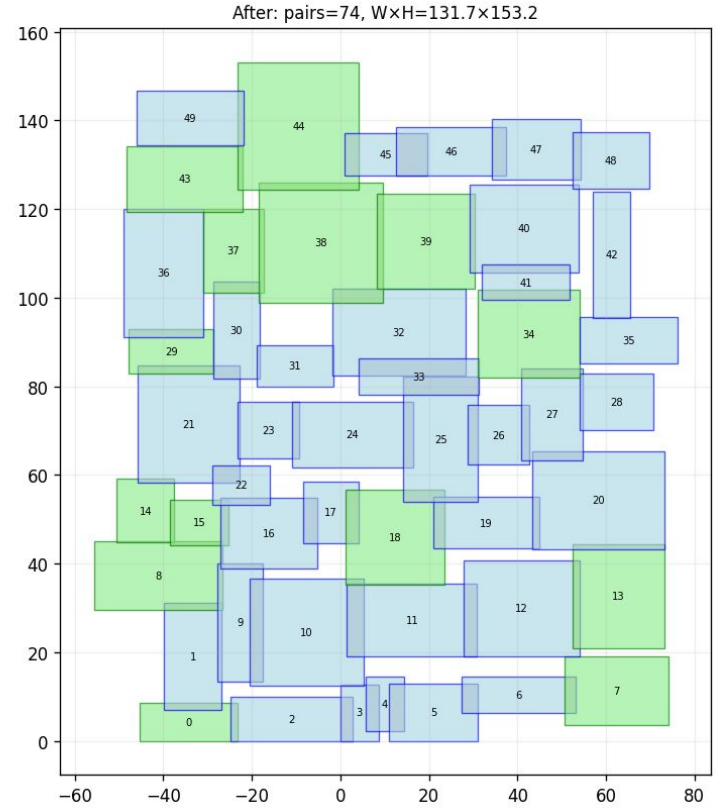
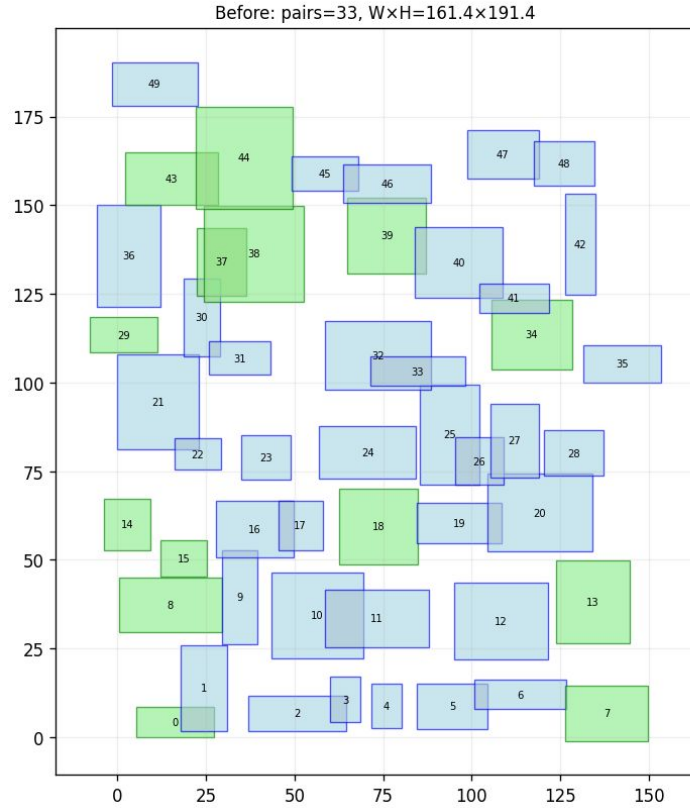
Legalization: Force directed

- 縮小邊框：
把往內靠攏的力拿掉，換成一個框框在收縮以及偵測空白點並且提供吸引力，並且有新增旋轉的機率。
- 最後，當框框縮小至1.03%所有block面積和時讓框框在訓練的最後30%反覆地放大縮小五次，範圍落在1.03%到1.13%，並且將旋轉的條件從原本的訓練前90%線性遞減改成由面積決定：初始框框面積到1.10%所有block面積和的線性遞減。

Legalization: test_01.csv



Legalization: test_00.csv





Legalization: Force directed

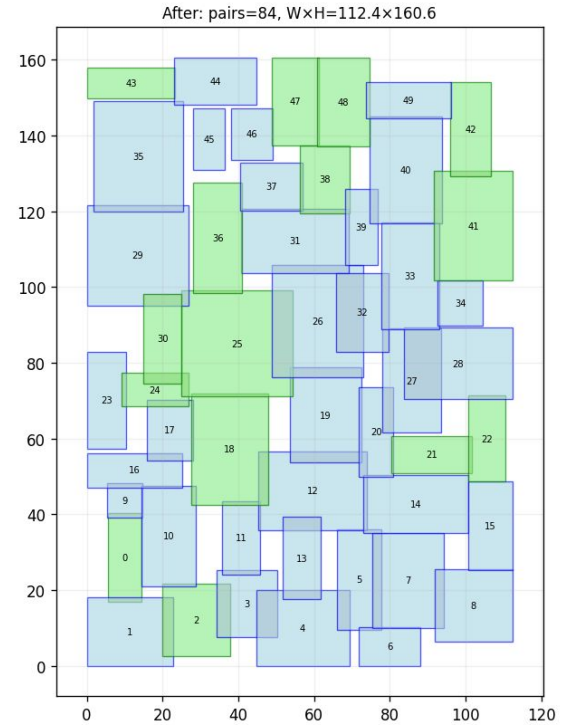
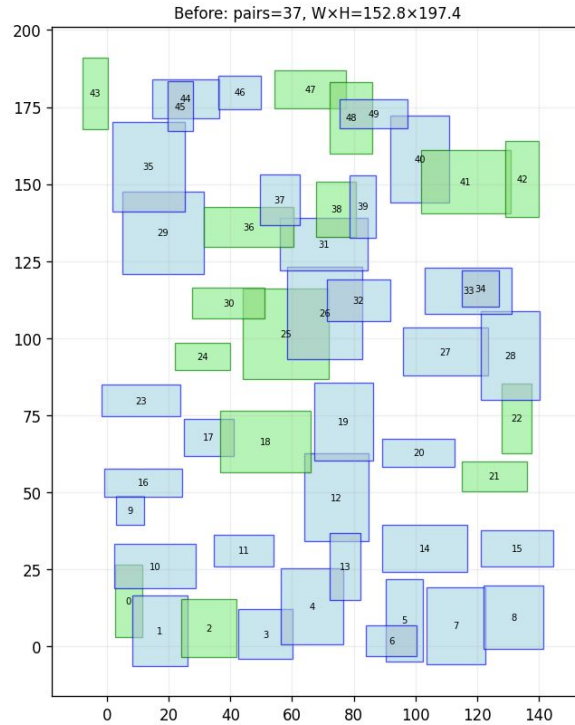
- 把frame wall改成硬性
- 把決定收縮的速度比例調整成在0.2%到0%間偵測相鄰的blocks, 這些blocks垂直向外的受力總和來決定, 而不是目前的推開。
- 再新增一個力, 叫做牆壁對block的吸引力, 跟block的距離平方成反比, 請注意上面的那個決定收縮的速度比例考慮的力要額外不考慮這個牆壁的吸引力, 考慮所有其他的力。

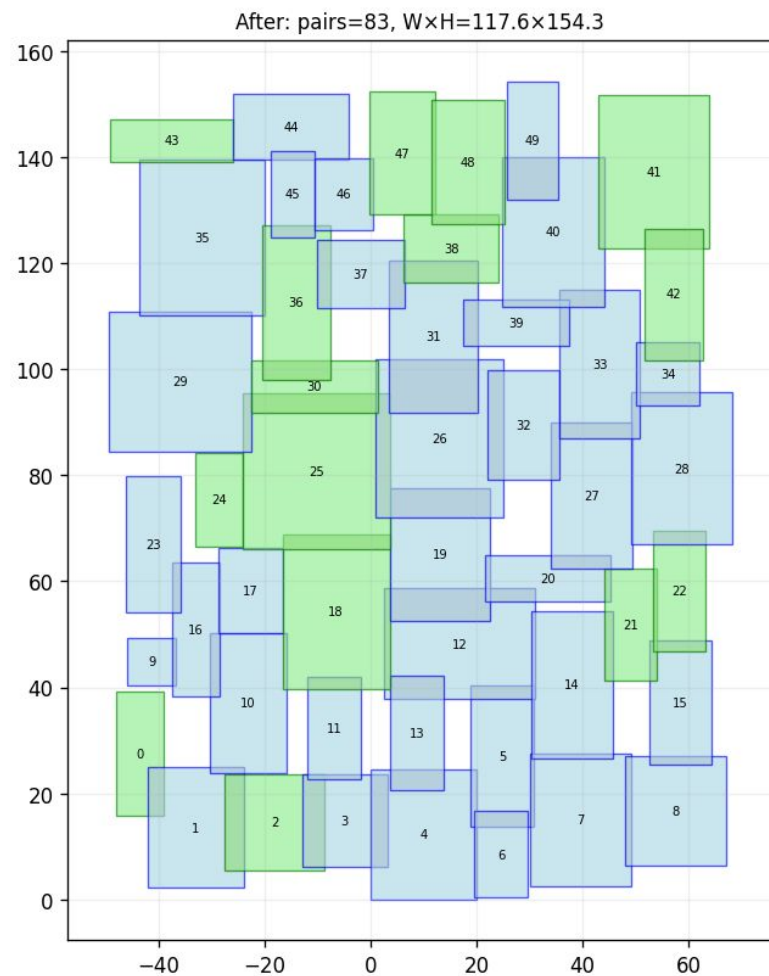
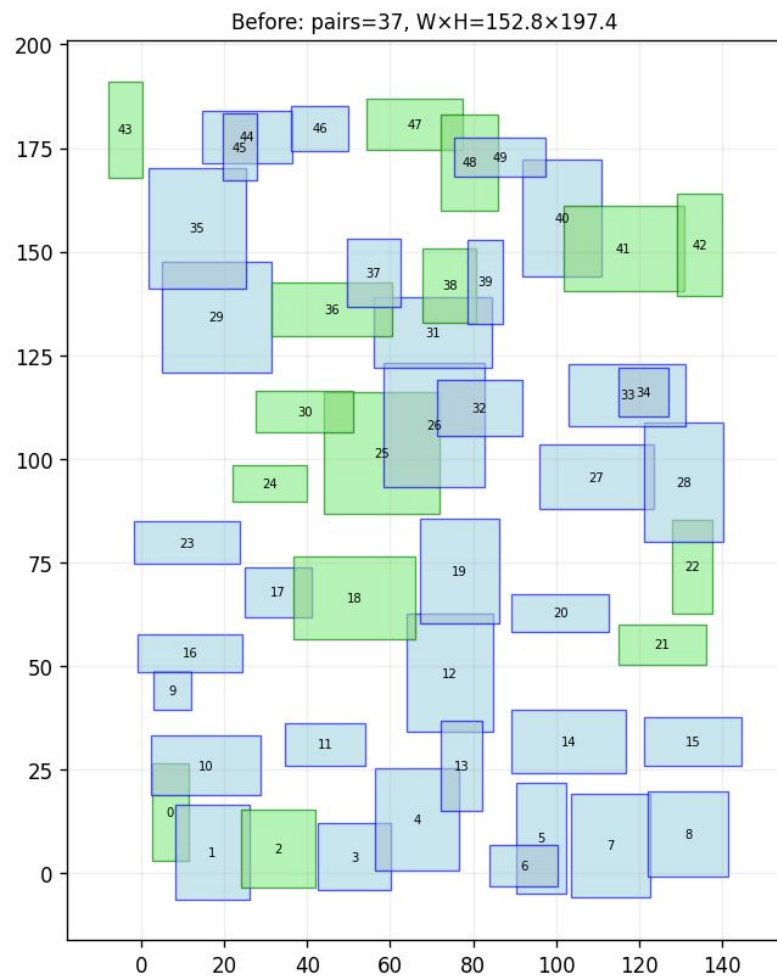


Legalization: Force directed

- 新增旋轉並設定cool down
- 加入另一stage: 不再shrink

Legalization: test_01.csv



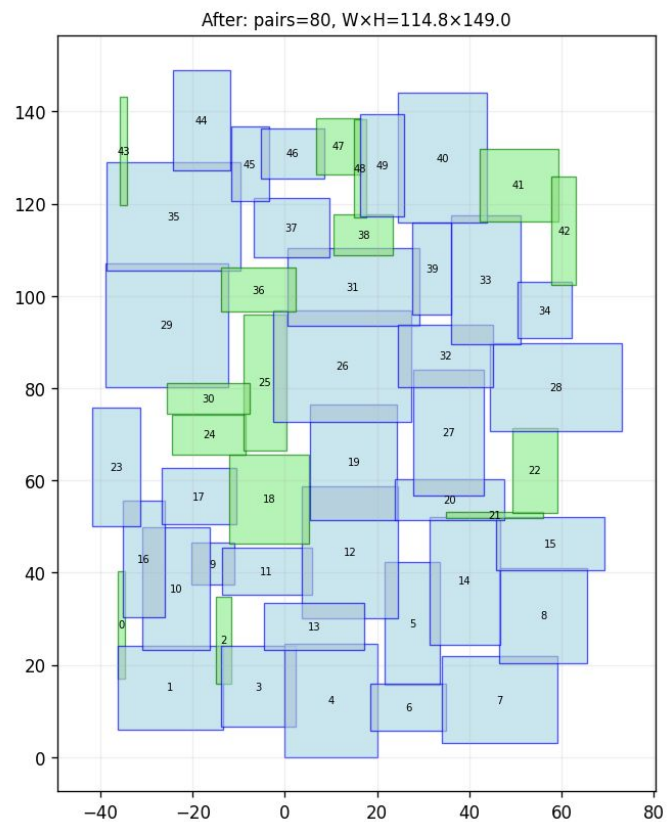
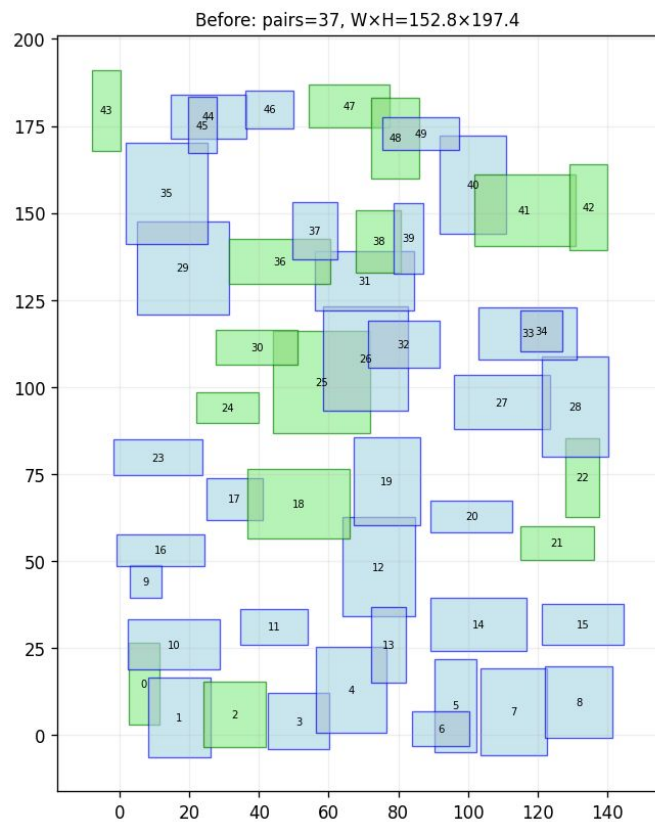




Legalization: Force directed

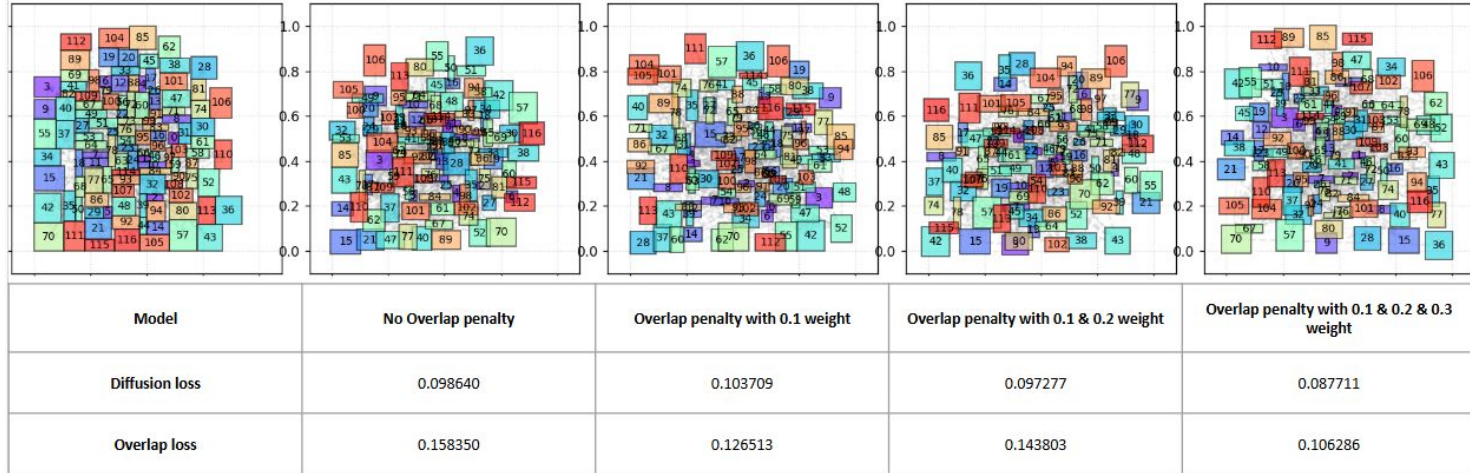
- 加入soft blocks
- 使用力導向推邊

Legalization: test_01.csv



Recall

Diffusion Model for Floorplanning - Predict



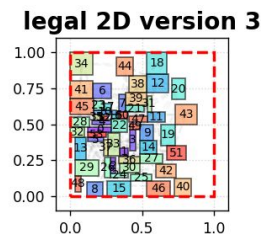
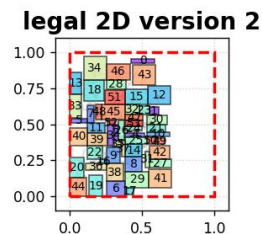
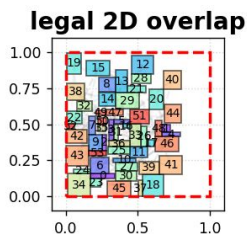
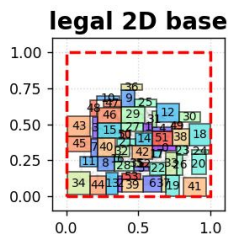
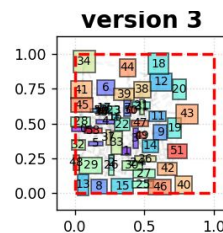
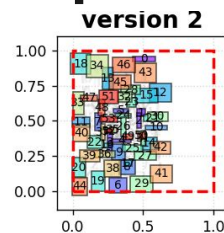
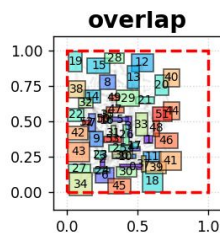
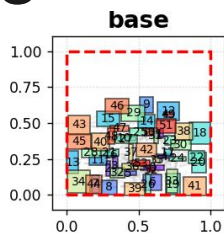


Legalization - Greedy

- Big blocks place first
- When placing smaller block
 - No overlap -> place it directly
 - Overlap -> Find a new space
- Finding space
 - Drift penalty -> Legal 2D
 - Cost driven -> Legal 2D cost driven

base: Only MSE loss
 overlap: with 0.1 weight overlap loss
 v2: with 0.1 & 0.2 weight overlap loss
 v3: with 0.1 & 0.2 & 0.3 weight overlap loss

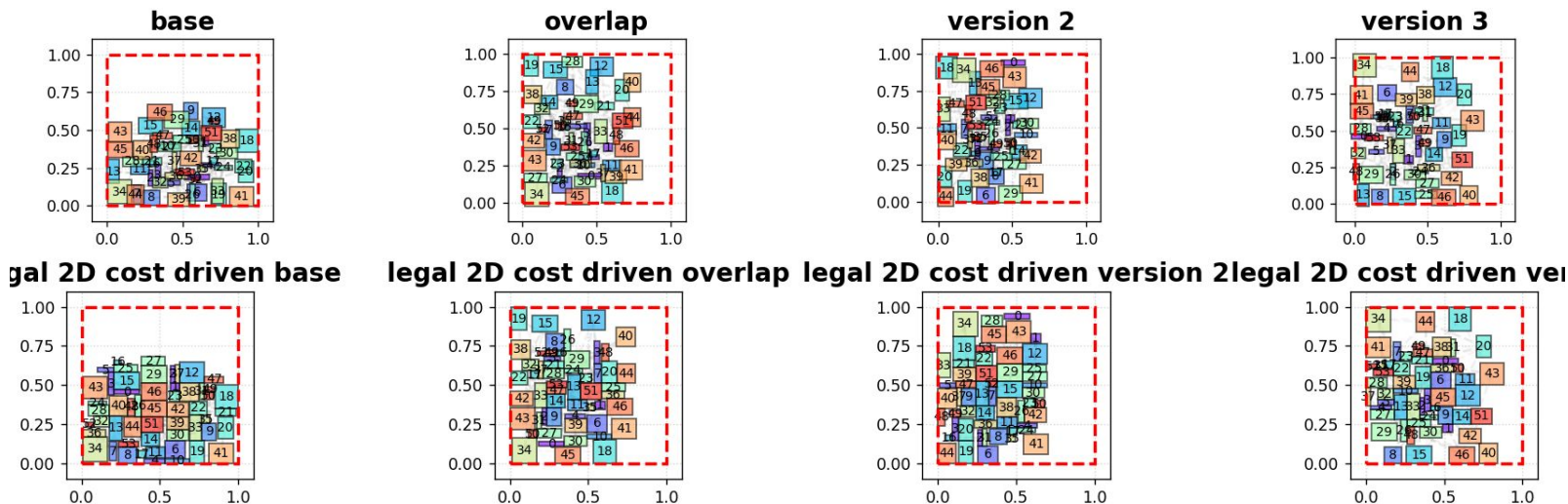
Legalization Result - Dataset Input



```
base cost: 20.192566, total_area: 0.776796, HPWL: 39.608337
overlap cost: 20.597586, total_area: 0.805105, HPWL: 40.390068
version 2 cost: 17.954323, total_area: 0.685149, HPWL: 35.223495
version 3 cost: 18.603754, total_area: 0.883627, HPWL: 36.323879
```

base: Only MSE loss
 overlap: with 0.1 weight overlap loss
 v2: with 0.1 & 0.2 weight overlap loss
 v3: with 0.1 & 0.2 & 0.3 weight overlap loss

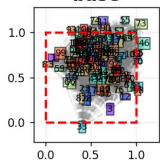
Legalization Result - Dataset Input



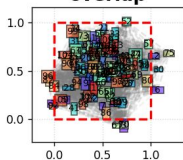
base cost: 19.134407, total_area: 0.679130, HPWL: 37.589684
 overlap cost: 18.098642, total_area: 0.805105, HPWL: 35.392181
 version 2 cost: 17.861176, total_area: 0.685149, HPWL: 35.037201
 version 3 cost: 17.729710, total_area: 0.883627, HPWL: 34.575790

Legalization Result - New Input

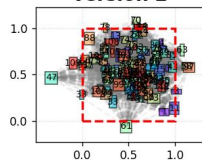
base



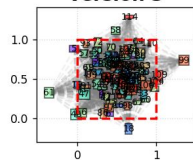
overlap



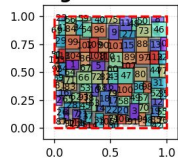
version 2



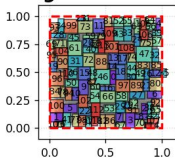
version 3



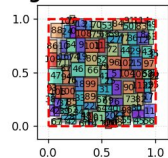
legal 2D base



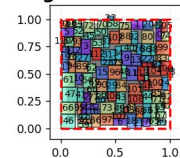
legal 2D overlap



legal 2D version 2

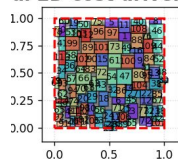


legal 2D version 3

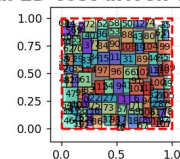
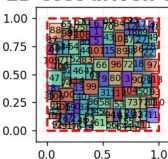
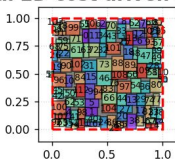


```
base cost: 1109.814941, total_area: 0.998888, HPWL: 2218.631104
overlap cost: 1107.090942, total_area: 1.044599, HPWL: 2213.137207
version 2 cost: 1124.875000, total_area: 1.071146, HPWL: 2248.678955
version 3 cost: 1106.186401, total_area: 1.058319, HPWL: 2211.314453
base cost: 1107.721558, total_area: 1.079587, HPWL: 2214.363525
overlap cost: 1146.325073, total_area: 1.080395, HPWL: 2291.569824
version 2 cost: 1117.629028, total_area: 1.000000, HPWL: 2234.258057
version 3 cost: 1120.810303, total_area: 1.018502, HPWL: 2240.602051
```

al 2D cost driven base

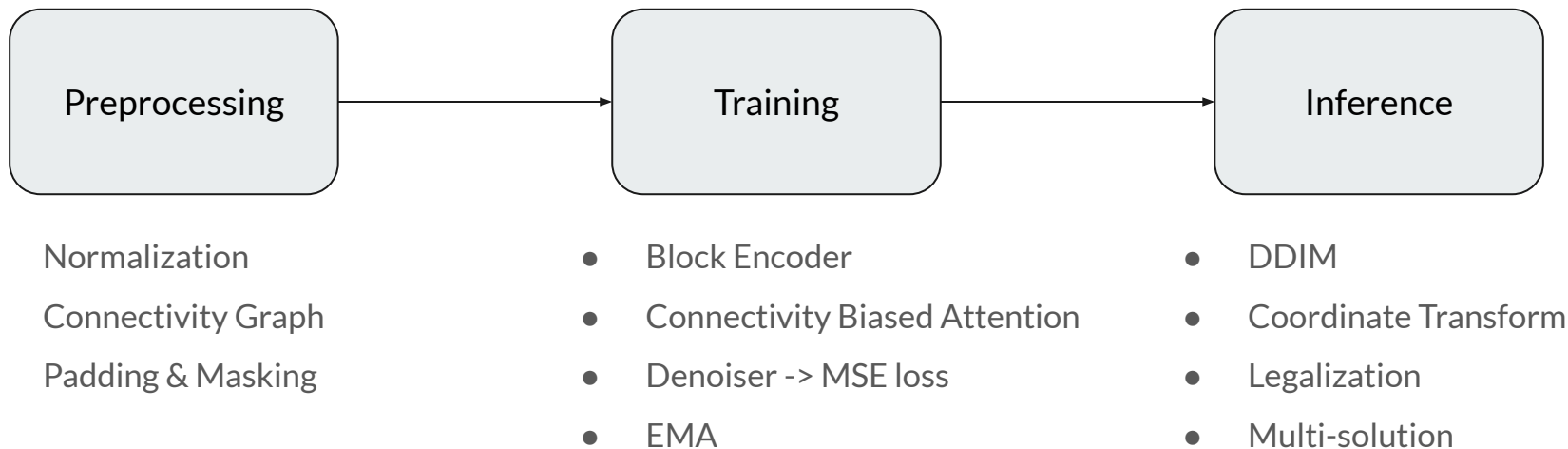


legal 2D cost driven overlap legal 2D cost driven version 2 legal 2D cost driven version 3





New Model





New Model - Preprocessing

- **Normalization:** Compress coordinate to $[0, 1]$ & calculate **log aspect ratio** ($\ln(w/h)$)
- **Connectivity Graph:** Generate **adj matrix** for training
- **Padding & Masking:** Adding 0 and boolean masks to fit fixed tensor size
- **Config:**
 - Dataset: 50 k (ICCAD)
 - Epochs: 30



New Model - Training

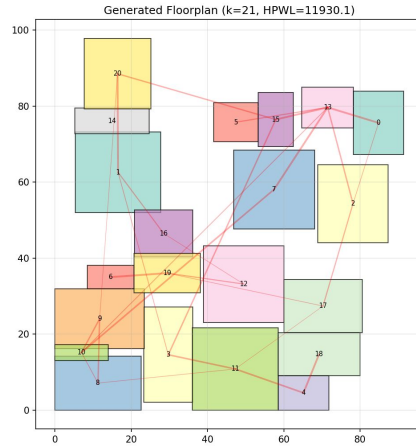
- **Block Encoder:** Encode area & topology into **condition embeddings**
- **Connectivity Biased Attention:** $\text{softmax}(\frac{QK^T}{\sqrt{d_k}} + \text{conn_bias})V$ -> Inject **physical connectivity** as prior knowledge
- **Denoiser:** Predict injected noise at random timestep t -> **MSE Loss**
- **EMA:** Maintain **shadow weights** for stable and higher-quality inference



New Model - Inference

- **DDIM:** Accelerate sampling process (1000 steps -> 50 steps)
 - No noise adding when inference
 - Predict the final x_0 and aim to it
- **Coordinate Transform:** Transform the compressed coordinate back
- **Legalization:** Iteratively resolve block overlaps via **greedy push heuristic**
- **Multi-solution:** Generate multi-solution and choose the **least overlap one**

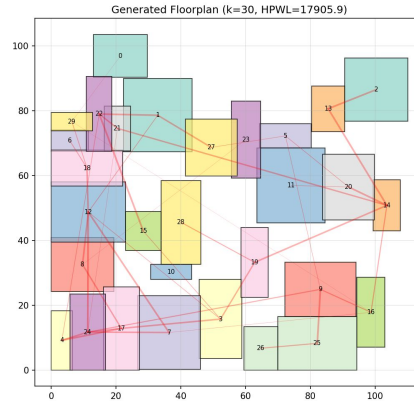
New Model - Result



Floorplan Evaluation Report

```

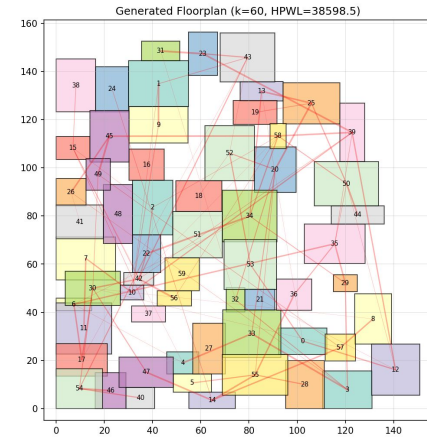
Blocks:      21
HPWL:       11930.14
Bbox area:   8941.64
Total overlap: 69.0540
Overlap pairs: 15
Max area err: 0.000000 (threshold: 0.01)
Mean area err: 0.000000
Feasible:    NO
  
```



Floorplan Evaluation Report

```

Blocks:      30
HPWL:       17905.95
Bbox area:   11379.26
Total overlap: 286.6548
Overlap pairs: 26
Max area err: 0.000000 (threshold: 0.01)
Mean area err: 0.000000
Feasible:    NO
  
```

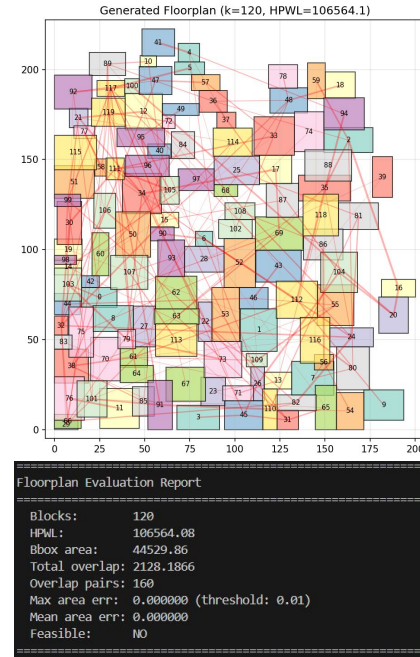
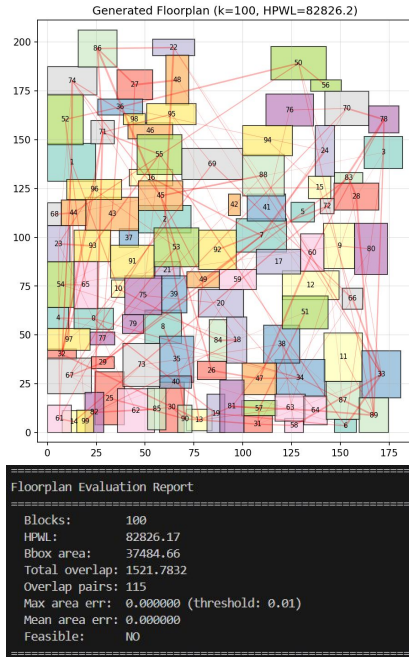


Floorplan Evaluation Report

```

Blocks:      60
HPWL:       38598.45
Bbox area:   23587.34
Total overlap: 663.5975
Overlap pairs: 72
Max area err: 0.000000 (threshold: 0.01)
Mean area err: 0.000000
Feasible:    NO
  
```

New Model - Result



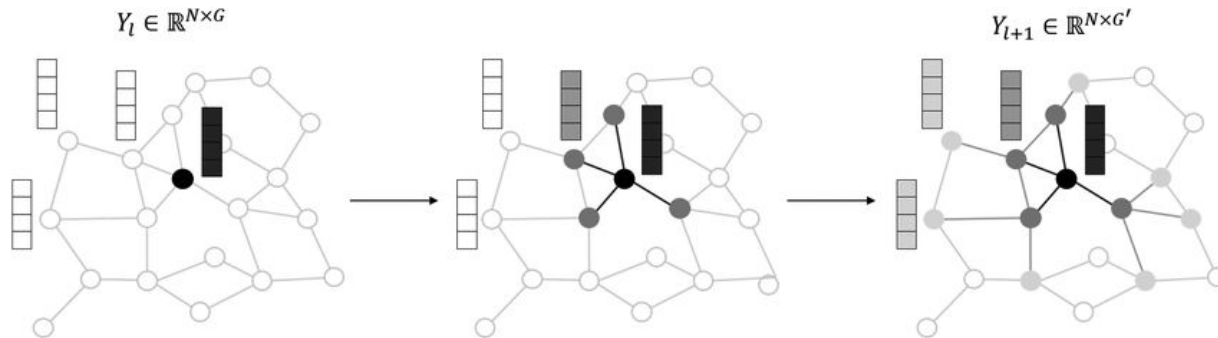


TODO

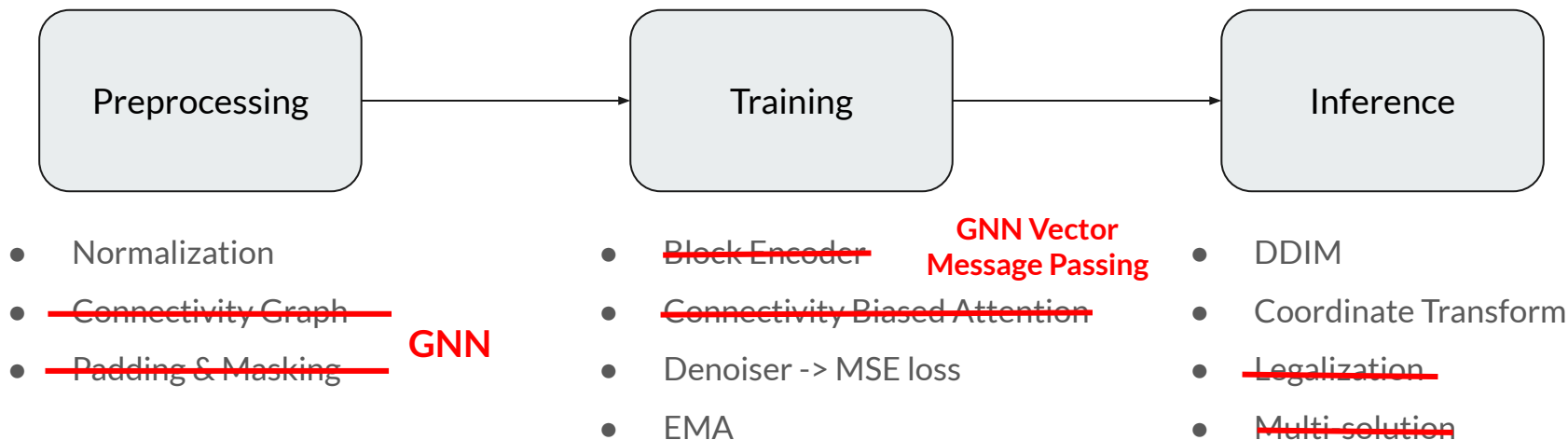
- Starting to consider the constraints
- Training with more datasets & epochs
- Legalization & Fine-tuning

GNN Attempt

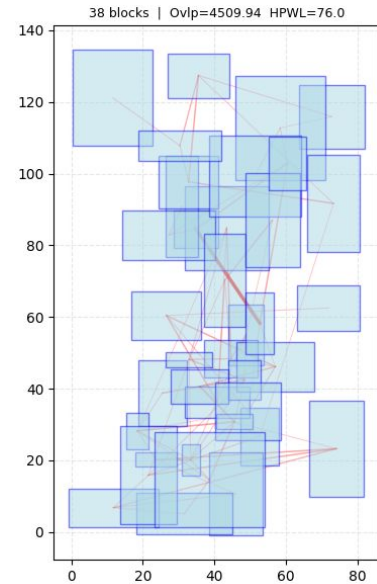
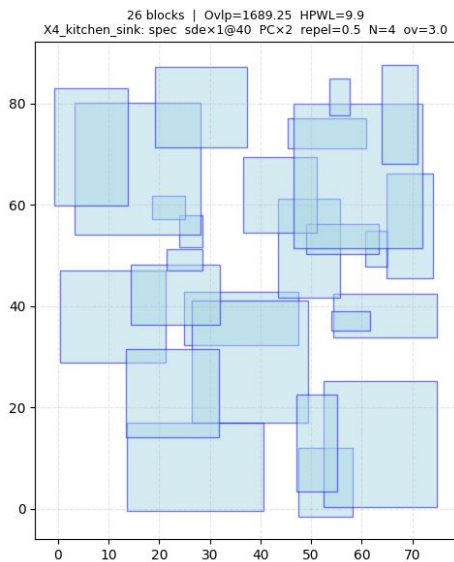
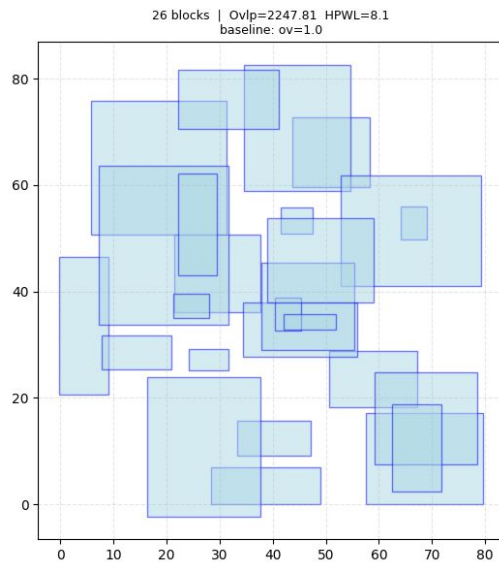
- Graph is an intuitive way to conceptualize floorplanning
- Use **Graph** to represent data instead of **Adjacent Matrix**



GNN Attempt - Implementation Detail

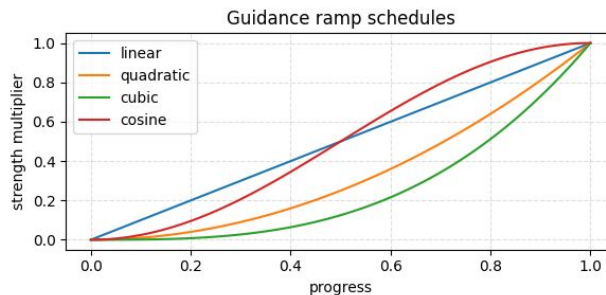


GNN Attempt - Results

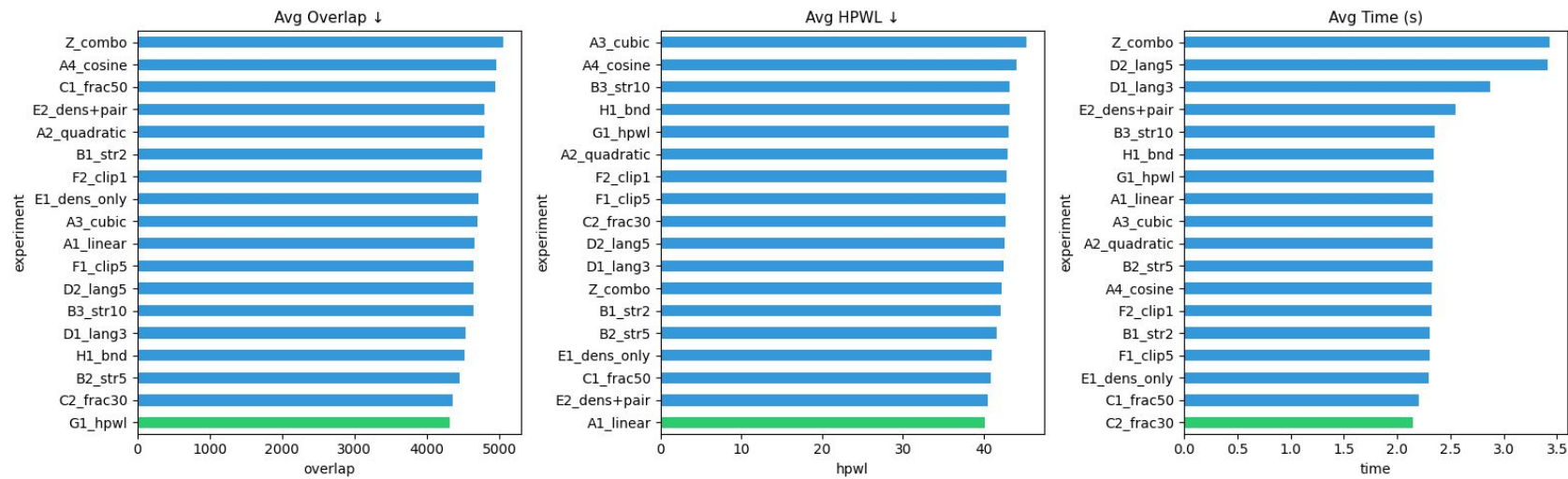


Experiment 1: Schedule & Guidance Strength

#	Strategy	What it does
A	Schedule shape	Linear / quadratic / cubic / cosine ramp for guidance strength
B	Guidance strength	Sweep <code>S_CONS_MAX</code> from 1.0 to 10.0
C	Guidance start fraction	Start guidance earlier (0.5) or later (0.9)
D	Multi-step guidance	Run K Langevin correction steps per timestep
E	Grid-based density	Replace $O(N^2)$ pairwise with $O(N+G^2)$ density field
F	Gradient clipping	Clip guidance gradient norm to prevent instability
G	HPWL co-guidance	Guide toward lower wirelength alongside overlap
H	Boundary guidance	Penalise blocks outside a target bounding box



Experiment 1: Schedule & Guidance Strength

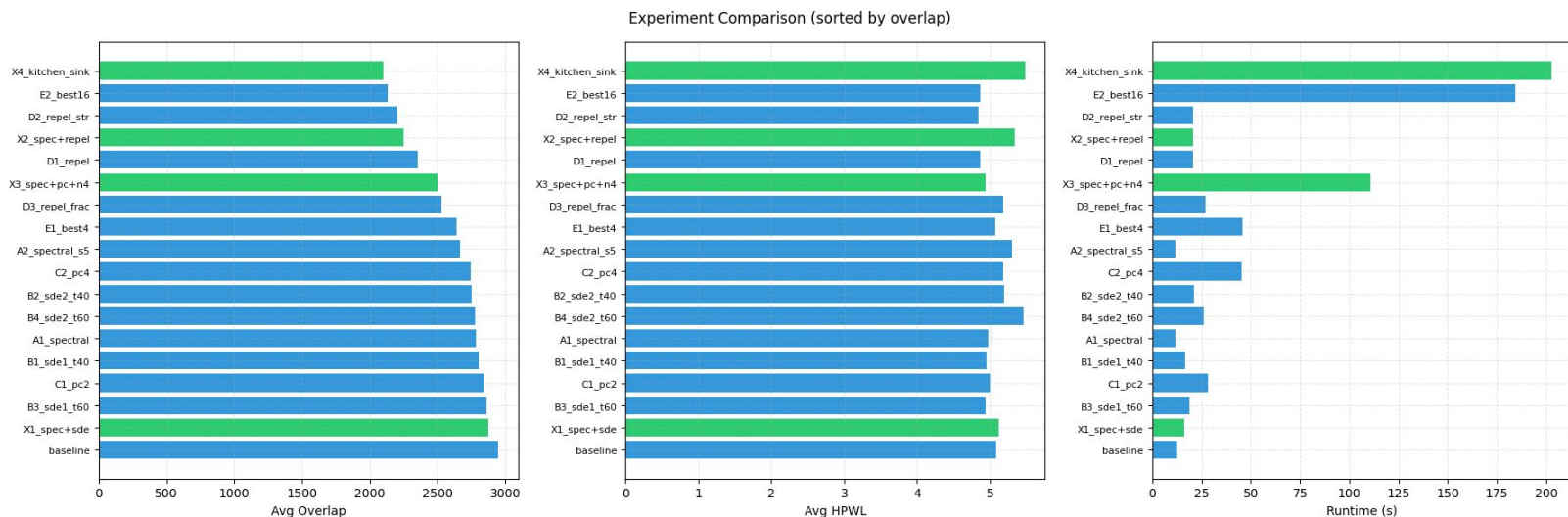




Experiment 2: Various Techniques

#	Method	Idea
A	Spectral Init	Warm-start from a quadratic placement solve
B	SDEdit (re-noise + re-denoise)	Give the model a second pass
C	Predictor-Corrector	Langevin steps that re-query the denoiser
D	Direct Repulsion Projection	Move blocks apart <i>after</i> each DDPM step
E	Best-of-N Sampling	Generate N layouts in parallel, keep best

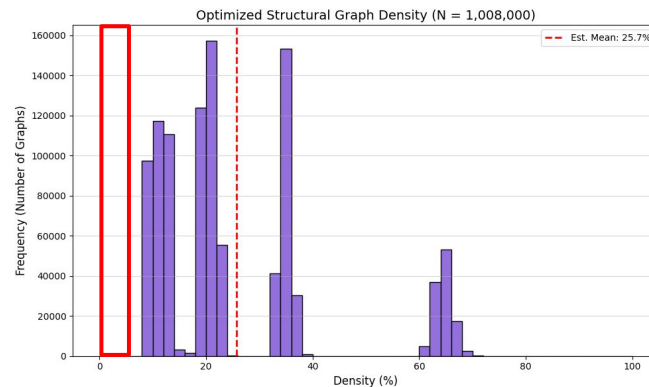
Experiment 2: Various Techniques



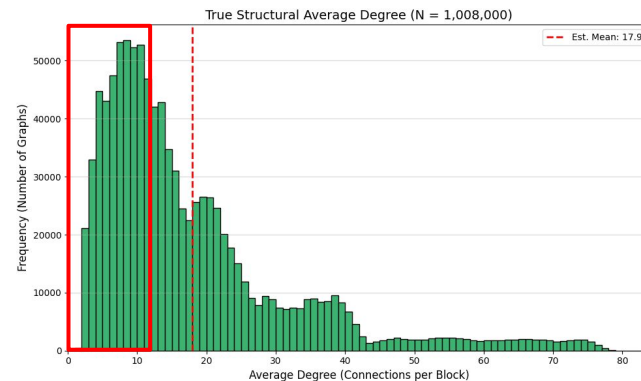
Probable Cause: Dense Graph

- Oversmoothing issue due to **dense graph**
- Solutions:
 - Top-K selection for edges
 - Graph Attention Network (GAT) learns to ignore the irrelevant weights
 - Graph Transformer Network

Reasonable range: <5%

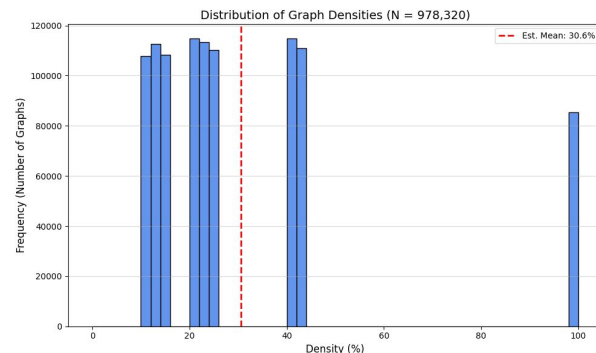


Reasonable range: 0~12



Accidental Discovery: Bidirectional Edges

- **Edge Representation:** (u, v, w) represents an edge from u to v with weight w .
- **Directionality of Edges in the Dataset:**
 - **Bidirectional:** Both (u, v, w) and (v, u, w) exist.
 - **Unidirectional:** Only (u, v, w) exists.
 - Both types exist in the dataset.



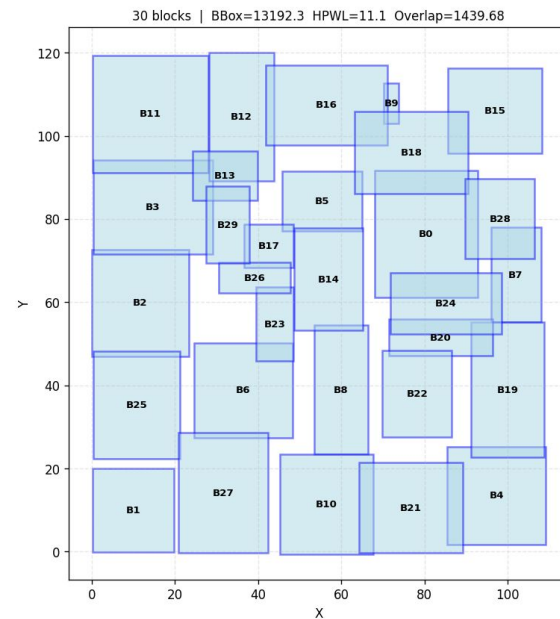
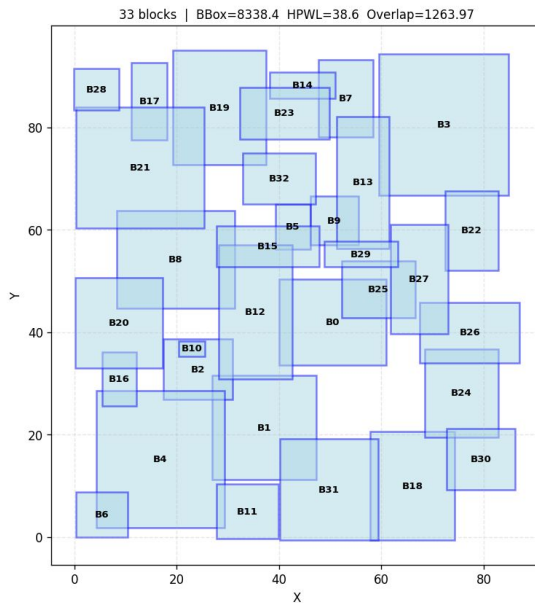
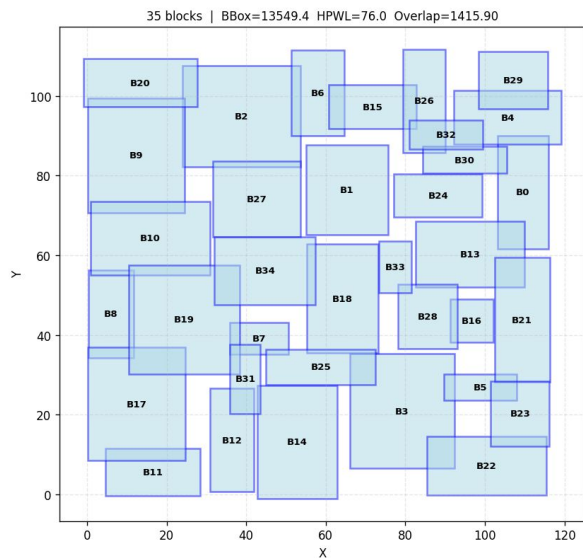
```
--- Anomaly Graph Found ---  
Nodes: 44, Edges: 948, Max Allowed: 946.0  
Total Edges: 948  
Self-Loops Found: 0  
Duplicate Edges Found: 0  
Bidirected Pairs Found: 340
```



GNN Attempt: TODO

- Train the network with overlap loss
- Experiment with different methods to deal with dense graphs

GNN Attempt: Latest Result





GNN Attempt: Methods & Parameters

- Implement fixes: **Max-K** selection, **overlap loss** with weight $\lambda = 0.1$, **deduplicate** the edges
- **Spectral Init**: Estimated initial position
- **Spatial edges**: For every block, we add edges to connect to 6 closest neighbors.
- Config:
 - Dataset: ICCAD 20~29 blocks (training: 78771, validation: 19693)
 - Epoch: 100



GNN Attempt: Next Steps

- Experiment with different overlap loss weight λ
- Testing viability on graphs with more blocks



Reference

[1] Image in p.14: https://www.researchgate.net/figure/A-graph-neural-network-GNN-layer-with-a-2-hop-neighbourhood-The-figures-from-left-to_fig3_374788593